

B M Tazbiul Hassan Anik

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EDUCATION

Ph.D. in Transportation Systems Engineering (CGPA: 3.94 / 4.00)

University of Central Florida

Exp. July 2025

Orlando, FL

- **Research:** Time-Series Forecasting, Causal Inference, Deep Learning, Bayesian Modeling, Experimentation.

B.Sc. in Civil Engineering (Major: Transportation)

Bangladesh University of Engineering and Technology

February 2021

Dhaka, Bangladesh

EXPERIENCE

Graduate Research Assistant

University of Central Florida

August 2022 – Present

Orlando, FL

- Applied Difference-in-Differences (DiD) to evaluate the impact of policy changes on pedestrian safety, identifying a **9% reduction in pedestrian-vehicle conflicts**.
- Implemented **Bayesian Negative Binomial modeling** for **crash frequency analysis**, capturing overdispersion and spatial variability to improve safety predictions.
- Built **Python-based ETL pipelines** with Docker and GitHub Actions, automating data ingestion and validation in AWS environments.
- Processed and analyzed **2.5+ million telematics data points** from OEMs using **Python and SQL**, enabling actionable insights for decision-making across county transportation networks.
- Designed A/B testing with survey data to assess real-time parking notifications, concluding an **increased intended shuttle adoption on campus by 21%**.
- Authored technical reports and collaborated with cross-functional teams, including state DOT and private agencies, to integrate machine learning models into production workflows.

Traffic Data Analyst

Accident Research Institute

July 2021 – June 2022

Dhaka, Bangladesh

- Queried and analyzed **17,000+ crash records** using **SQL**, identifying critical trends to inform traffic safety interventions.
- Applied **regression analysis** and **hypothesis testing** to evaluate safety outcomes with 95% confidence.
- Created visualizations (e.g., heatmaps, trend charts) using Matplotlib and Seaborn to communicate insights to policymakers and stakeholders.

NOTABLE PROJECTS AND PUBLICATIONS

Optimization of Safety KPIs | [Project Link](#) | Python, KPI

April 2024 – Present

- Streamlined the transformation of large-scale sensor-based event data into actionable performance metrics (KPIs) for intersection safety and efficiency analysis.
- Leading the development of an interactive dashboard for real-time and batch KPI monitoring, providing stakeholders with data-driven insights.

Incident Forecasting Model | [Paper Link](#) | Python, PyTorch

December 2024

- Built GAN and Transformer-based models for imbalanced time-series forecasting, improving recall by 24%.
- Deployed within the CPED system, enabling real-time traffic safety monitoring for the Florida Department of Transportation.

Adaptive Traffic Signal Adjustment | [Project Link](#) | Python, Causal ML

December 2024

- Applied Causal Forests to optimize traffic signal timing, reducing Red Light Running (RLR) probability by 7%.
- Accounted for temporal and spatial heterogeneity, ensuring adaptive, intersection-specific signal adjustments.

TECHNICAL SKILLS

Programming Languages

Python, SQL, R, HTML, CSS, JavaScript

Machine Learning & AI

Transformers, Random Forest, Causal Inference, Bayesian Modeling

Big Data & Cloud

AWS, Spark, Docker, CI/CD Pipelines, ETL Processes

Statistical Methods

A/B Testing, Factorial Design, Regression Analysis, Hypothesis Testing

Data Visualization

Tableau, Matplotlib, Seaborn, Plotly